Model Card — Product Sentiment Classifier

1. Model details

Field	Description
Model name	Product Sentiment Analysis (web app)
Task	Binary sentiment classification (positive vs. not positive) of FMCG product reviews
Model	TF-IDF features + Logistic Regression
Training data	FMCG Customer Sentiments (Kaggle): 3,500 reviews in 6 product categories; labels positive (1,415), negative (1,508), neutral (577)
Test data	1,500 records evaluated on the website (split method and random seed: to confirm)
Preprocessing	To confirm: lowercasing, stop-word handling, n-gram range, number of TF-IDF features
Deployment	Web app on Vercel that shows TF-IDF scores, the confusion matrix and a subgroup audit
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2. Intended use

- **Intended:** Flagging whether an English FMCG product review is positive, so a brand can triage customer feedback; classroom demonstration of a TF-IDF + Logistic Regression pipeline with a transparent audit.
- **Out of scope:** Decisions about individual customers or employees; reviews in other languages or code-mixed text; medical, legal or safety-critical content; separating neutral from negative reviews, because the confusion matrix is binary.

3. Overall performance

The model reaches 91.1% accuracy and 88.9% F1 on 1,500 evaluation records, beating an always-"not positive" baseline (59.8%) by 31.3 points.

Model	Accuracy	Precision	Recall	F1
Majority-class baseline	59.8%	—	0.0%	0.0%
TF-IDF + Logistic Regression	91.1%	89.6%	88.2%	88.9%

Confusion matrix	Predicted positive	Predicted not positive
Actual positive	532 true positives (35.5%)	71 false negatives (4.7%)
Actual not positive	62 false positives (4.1%)	835 true negatives (55.7%)

The model misses 71 positive reviews and wrongly flags 62 others, so its two error types are close to balanced. The baseline is derived from the matrix: 897 of the 1,500 records are not positive.

4. Performance across subgroups

Key audit finding: Medium-length reviews (21–49 words) score lowest of the three length groups at 84.4% F1, 7.7 points below detailed reviews (92.1%). The Baby Care category (85.8% F1) also trails the overall 88.9%.

Subgroup	How it was defined	Test size (n)	F1	Website flag
Short reviews	20 words or fewer	888	89.6%	Robust
Medium reviews	21–49 words	331	84.4%	Disparity flagged
Detailed reviews	50 words or more	281	92.1%	Highest

5. Explainability check

Method 1 — Top features (Logistic Regression coefficients). Provisional: these weights come from a re-run of TF-IDF (unigrams + bigrams) with Logistic Regression (C = 1.0) on the same dataset, predicting positive vs. not positive. Replace them with the website's own values if they differ.

Top positive features (weight)	Top negative features (weight)
highly (+2.6)	not (−3.5)
loved (+1.9)	quite (−1.6)

Top positive features (weight)	Top negative features (weight)
five stars (+1.9)	terrible (−1.5)
would buy (+1.8)	would not (−1.5)
highly recommended (+1.7)	waste of (−1.3)

The negation cue "not" carries the strongest weight in either direction, and the bigrams "not worth" (−1.08) and "not good" (−0.58) are learned as negative, while "good" alone is positive (+0.97). Function words such as "the" and "was" also carry negative weight and are left out of the list above.

Method 2 — Word contributions on a misclassified review (coefficient × TF-IDF value, from the same provisional re-run).

- Review: *"Good effectiveness overall, but not impressed with the smell. Would buy again."* True label: positive; predicted: not positive (p(positive) = 0.41).
- "would buy" (+0.38) and "impressed" (+0.16) push towards positive, while "not" (−0.37) and "but not" (−0.15) push away.
- **Interpretation:** this is a mixed review. The praise and the complaint nearly cancel, so the probability lands just below 0.5. It is the mixed-sentiment case this project set out to audit.

6. Limitations and ethical considerations

- **Main limitation:** TF-IDF scores weighted words and cannot read full context, so negation and mixed opinions are the main risk. The weakest measured group is medium-length reviews (84.4% F1).
- **Binary framing:** The dataset has three labels, but the website reports a binary confusion matrix, so the 577 neutral reviews are not scored as their own class.
- **Repeated reviews:** Only 2,618 of the 3,500 review texts are distinct, so identical reviews can fall in both training and test data and inflate scores.
- **Narrow data:** English reviews from six FMCG categories with a median length of 13 words; results may not carry over to other products or longer reviews.
- **Small audit samples:** Each subgroup holds 258 to 888 records, so a gap of a few points may be within noise.

7. Recommendations

- Remove duplicate review texts before splitting the data, then re-run the evaluation.

- Add a negation subgroup and a mixed-sentiment subgroup to the audit; 1,678 reviews contain a negation word such as "not" or "never".
- Benchmark against a Transformer model (for example a fine-tuned DistilBERT) and repeat the same length and category audit.
- Extend the classifier to three classes so neutral reviews are scored.
- Send low-confidence predictions (probability between 0.4 and 0.6) to a human reviewer.